

Housing & Disaster Trends

Figure 1: Housing & Disaster Trends

Lag Sensitivity to Mortgage Rates

Figure 2: Lag Sensitivity to Mortgage Rates

QM 2023 Capstone Project: Milestone 4 Presentation (Visual Focus)

Slide 1: Title

- How do repeated natural disasters affect local housing price growth and volatility in high-risk counties?
 - Team: Brice, Ricky, Zach
-

Slide 2: Key Finding (Visual)

- Main result: Disasters have little effect on housing price growth.
-

Slide 3: Macro Drivers Dominate

- Mortgage rates and macro factors matter more than disasters.
-

Slide 4: Model Diagnostics

- Model is statistically clean but noisy.
-

Slide 5: Forecasting Results

- ARIMA and ML models improve prediction, but error remains.
-

Residuals vs Fitted

Figure 3: Residuals vs Fitted

ARIMA Forecast

Figure 4: ARIMA Forecast

ML Predictions Comparison

Figure 5: ML Predictions Comparison

Slide 6: Machine Learning Comparison

- Random forest vs OLS: ML is slightly better.
-

Slide 7: Summary & Recommendation

- Focus on rate-sensitive markets, not disaster history.
 - Use disaster as a secondary risk screen.
-

Slide 8: Q&A

- Questions?